H1

ANDERSON SERANGOON JUNIOR COLLEGE
HIGHER 1

2021 JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME	**MARK SCHEME**

CLASS	2	1	/			INDEX NUMBER	

BIOLOGY 8876/02

Paper 2 Structured and Free Response Questions

15 September 2021
Wednesday

Candidates answer on the Question Paper.
No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Total (P1)	**/ 30**
Section A	
1	
2	
3	
4	
5	
Section B	
6 or 7	
Total (P2)	**/ 60**
100%	
Grade	

Section A

Answer **all** the questions.

1 Microorganisms and their associated enzymes are economically important, and are widely used in various industries.

(a) For a commercial application using an enzyme, the progress of the enzyme-catalysed reaction needs to be studied.

Outline how the progress of an enzyme-catalysed reaction can be investigated experimentally.

1. by measuring the **rate of substrate disappearance/ rate of product formation (appearance)**

[any 2]

2. *ref. to* obtaining samples at **regular/ fixed time intervals**
3. to **determine substrate concentration/ product concentration**
4. *ref. to* controlled/ constant variables, e.g. **constant pH**, **constant temperature**
5. plot a **graph** of dependent variable (y-axis) against time (x-axis)

[3]

The enzyme lactase is commercially extracted from the unicellular fungus *Kluyveromyces lactis*, which is naturally found in dairy products. Lactase catalyses the breakdown of lactose, a sugar found in milk, and is used to produce lactose-free milk.

The reaction catalysed by lactase is summarised in Fig. 1.1.

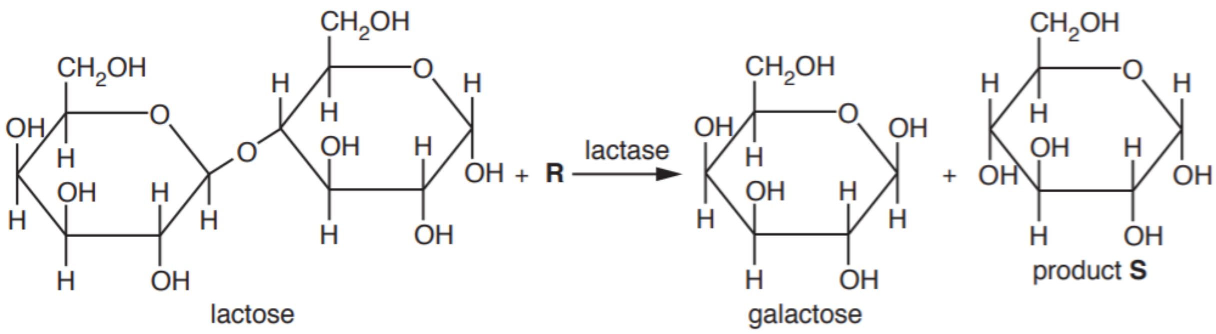

Fig. 1.1

(b) Describe the reaction that is catalysed by lactase. Use Fig. 1.1 to help you.

In your answer, identify **R** and product **S**.

[any 2 MPs for 1 mark]

1. The **glycosidic bond** is broken
2. in a **hydrolysis** reaction
3. with the addition of a **water** molecule (R)
4. to yield **galactose** and **α-glucose** (S)
5. AVP, e.g. description of **induced fit** or **lock and key** hypothesis, **disaccharide** hydrolysed to two **monosaccharides**

[2]

In the syrup industry, the starch extracted from barley seeds (barley starch) is used in the production of sugar syrups. One key enzyme synthesised by the barley seed is α-amylase, which breaks down stored starch during seedling formation.

Fig. 1.2 is a graph showing how the activity of α-amylase extracted from barley seeds changes as the temperature increases from 10 °C to 66 °C.

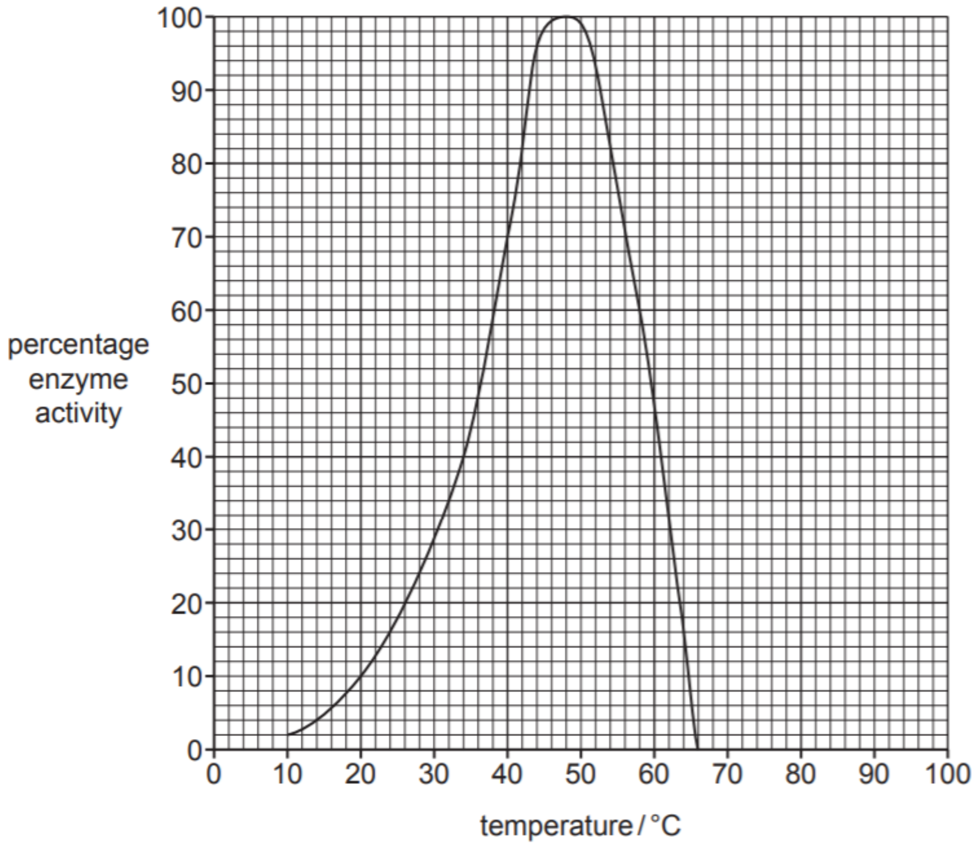

Fig. 1.2

(c) Explain the effect of temperature on the activity of α-amylase extracted from barley seeds, as shown in Fig. 1.2.

[any 3]

1. Increase in temperature **increases kinetic energy** of **substrate**/ **starch** molecules and **enzyme**/ **α-amylase** molecules
2. leading to an **increase** in the **rate** of **effective/ successful collisions** between enzyme and substrate/ **enzyme–substrate complex formation**/ **substrate** binding to **active site**
   OR
   at **48 °C** *(Accept: 47 or 49)*, *ref. to* **optimum** temperature/ **maximum enzyme activity**/ **active sites binding substrate at maximum capacity**/ AW ;
3. (at temperatures higher than optimum/ 48 °C), ref. to **denaturation**/ **loss of shape** of **active site** (and decrease in activity)
4. AVP, e.g. weak/ hydrogen/ ionic bonds break *(and ref. to denaturation)*

[3]

(d) When producing sugar syrups, there are advantages in using enzymes extracted from microorganisms, rather than enzymes extracted from barley seeds.

For example, some enzymes extracted from microorganisms are heat stable. Heat-stable enzymes are used to increase productivity because the reactions can be carried out at higher temperatures.

Sketch on Fig. 1.2 the curve that would be obtained using α-amylase enzyme that is heat stable. [1]

- optimum/ peak/ max activity at 100% and to right of original
- line beyond optimum must be to right and reach vertical axis at 50% or below/ reach horizontal axis

Microorganisms, specifically bacteria, are also used in the mining industry.

When gold is associated with mineral ores such as iron sulfide, the sulfides must be oxidised to release the gold particles.

Since the mid 1990s, gold has been extracted from such ores by bioleaching. Bioleaching involves the use of suitable bacteria such as *Acidithiobacillus ferrooxidans*, which oxidise iron sulfide to soluble iron sulfate, releasing Fe^{3+} and SO_4^{2-} ions.

Gold-bearing sulfide ores often contain arsenic, which is potentially toxic to the bacteria used in bioleaching. However, arsenic-resistant strains of *A. ferrooxidans* have been found in some mines.

The activity of two strains of the bacterium, in the presence and absence of arsenic ions, is shown in Table 1.1.

Table 1.1

	oxidation rate of iron in the ore / mg dm^{-3} h^{-1}	
strain of *A. ferrooxidans*	arsenic ions absent	arsenic ions present
1	16	15
2	48	47

(e) Describe the results shown in Table 1.1 **and** explain the role of natural selection in the evolution of arsenic-resistant bacteria.

[any 4]

1. Both strains give **similar rate with and without arsenic ions**
2. Both strains are **arsenic-resistant**
3. Strain 2 has a **higher oxidation rate**/ is **more active** (than strain 1)

[max 3]

4. arsenic acts as a **selective agent**/ exerts a **selection pressure**
5. spontaneous **mutation** produces resistant bacteria
6. Resistant bacteria is able to **survive** and **resistant allele** is **passed on** daughter cells
7. **Frequency of allele increases** in **population**/ **gene pool** (of population) over time

[4]

[Total: 13]

2 Fig. 2.1 shows the life cycle of the water flea (*Daphnia sp.*), which lives in various aquatic environments ranging from acidic swamps to freshwater lakes.

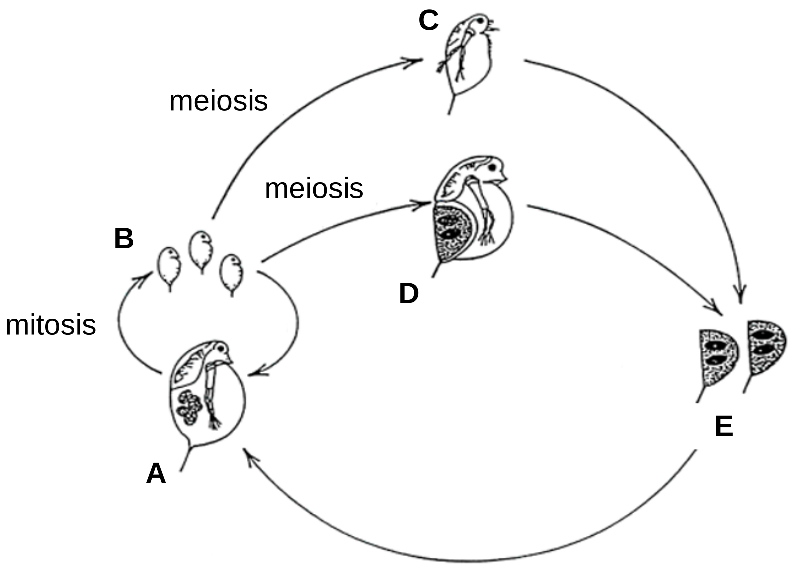

Fig. 2.1

- In favourable conditions, all the animals in a population are females (**A**).
- These females produce eggs by mitosis, which develop into young females (**B**) without being fertilised.
- In unfavourable conditions, eggs are produced by meiosis and develop directly without fertilisation into either males (**C**) or females (**D**).
- The eggs produced by females (**D**) are fertilised by sperm from the males (**C**), and then released in a protective egg case (**E**) which enables them to survive unfavourable conditions. When favourable conditions return, these fertilised eggs develop back into females (**A**).

(a) Explain why the eggs from **D** and the sperm from **C** must be produced by mitosis.

[any 2]

1. **C and D** are **haploid** (since they are formed from unfertilised eggs)
2. Mitosis ensures that the **haploid chromosome number** is preserved in the gametes/ eggs and sperm are **haploid**
3. so that fusion of haploid sperm and haploid egg **restores the original/ diploid chromosome number**

[2]

(b) Explain why females (**A**) that developed from fertilised eggs (**E**) are genetically different from one other.

1. Fertilised eggs E (that gave rise to females A) are the products of **<u>random</u> fertilisation**/ **<u>random</u> fusion of gametes**
2. **Gametes/ egg and sperm** that fuse are in turn **genetically different**, since they are the products of **mitosis from genetically different C and D** which are produced by **meiosis from adult A**
3. Meiosis in A leads to genetically different C and D due to **crossing over** between non-sister chromatids during **prophase I/ independent assortment** of chromosomes during **metaphase I (and II)**

[3]

(c) Biologists studied a population of water fleas in Felbrigg Hall, a shallow lake in England. The immediate region where the lake is located had experienced a threefold increase in the number of heat waves, and an average temperature rise of 1.15°C since the 1960s.

The biologists reported that the water flea population has **not** declined since the 1960s, and has since adapted to the warmer lake conditions.

Using biological knowledge and the information provided including Fig. 2.1, comment on how the water flea population in Felbrigg Hall is likely to be different today, compared to the water flea population in the 1960s.

1. *ref. to* **higher proportion of males**/ **more eggs**/ **fewer females (A) undergoing mitosis** as reproduction

[any 1]

2. *ref. to* **enzymes** in water fleas today having a **higher optimum temperature**
3. AVP, e.g. increased metabolism/ faster rate of development/ shortened life cycles *(flea population has adapted to higher temperatures)*

[2]

[Total: 7]

3 The mitochondrion is an important organelle found in eukaryotic cells.

(a) A student made the following statement:

'Without the mitochondrion, cellular respiration cannot occur.'

Comment on the accuracy of the statement above.

The statement is accurate only to some extent/ is inaccurate because

1. **anaerobic** respiration, which comprises **glycolysis** and **fermentation**, does not involve the mitochondrion since it occurs solely in the **cytosol**/ **cytoplasm**
2. Main stages of **aerobic** respiration, namely the **link reaction**, **Krebs cycle** and **oxidative phosphorylation**, **occur in the mitochondrion**/ **link reaction and Krebs cycle** occur in the **mitochondrial matrix**, while **oxidative phosphorylation** occurs on the **inner mitochondrial membrane**.

[2]

The mitochondrion contains its own DNA. Mitochondrial DNA (mtDNA) is a double-stranded molecule that consists of approximately 33,000 nucleotides.

(b) The two strands of mtDNA differ in their base composition, thus allowing both strands to be separated to give a heavy and a light strand by density centrifugation.

Suggest how the heavy strand differs from the light strand.

1. The heavy strand contains more **purines**/ **guanine** (accept: adenine) than the light strand.

[1]

The replication of mtDNA involves the Twinkle protein, which has both helicase and primase activity.

In humans, the Twinkle protein is encoded by the *TWNK* gene that is located in the long arm of chromosome 10.

(c) Fig. 3.1 is a simplified diagram of a human cell, which shows some key structures typically present.

On Fig. 3.1, use a labelling line and

(i) the letter **P** to label the precise location where transcription of the *TWNK* gene occurs. [1]

(ii) the letter **Q** to label the precise location where the Twinkle protein carries out its function. [1]

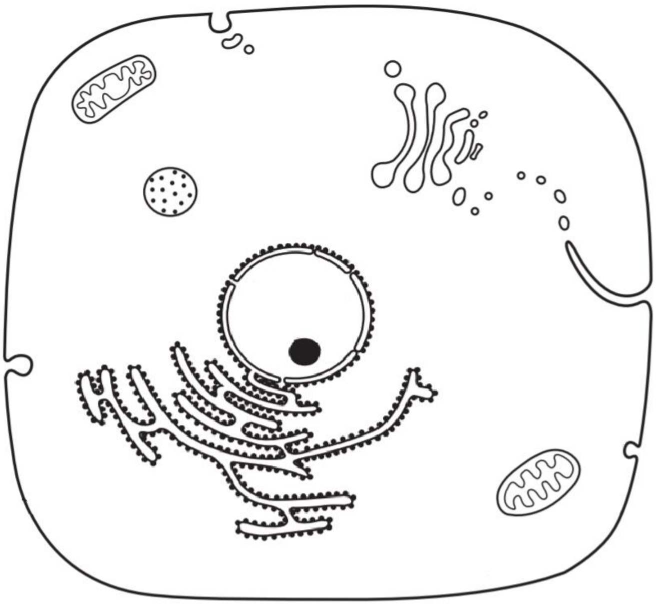

Fig. 3.1

1. labelling line with "P" indicating inside of the nucleus (Reject: labelling line touching nuclear envelope)
2. labelling line with "Q" indicating matrix of either mitochondria (Reject: labelling line touching outer mitochondrial membrane/ intermembrane space/ inner mitochondrial membrane)

(d) Explain the functions of the Twinkle protein during the replication of mtDNA.

[any 3]
helicase
1. **separates**/ **unwinds and unzips/ breaks hydrogen bonds** within double-stranded mtDNA
2. to form **single-stranded** DNA strands which serve as **template**

primase
3. catalyses the formation of an **<u>RNA</u> primer**
4. which provides a **3' OH** end for **DNA polymerase**

[3]

(e) The *TWNK* gene is also used to make the Twinky protein. While the function of this protein is unknown, some research suggest that it functions similarly to the Twinkle protein.

Explain how the single *TWNK* gene can be used to make two proteins.

1. via **alternative splicing** of **pre-mRNA**/ to **form mature mRNA**/ of **mRNA post-transcription**
2. where **different splice sites are cut at**/ only **some splice sites are cut at** such that **different combinations of exons** become spliced together/ result on mature mRNA. *(These mature mRNAs are then translated to form the two proteins)*

[2]

(f) There is broad evidence to indicate that the modern-day mitochondrion is the result of an ancient free-living bacterium which was integrated into an ancestral eukaryotic cell. This is known as the endosymbiont theory.

(i) One evidence for the endosymbiont theory is that the type of DNA possessed by both the mitochondrion and a bacterium is similar.

Apart from this similarity, identify another structural feature that is common to both the mitochondrion and a bacterium.

1. Presence of **70S ribosomes** in both structures

[1]

(ii) Another evidence for the endosymbiont theory is that the mitochondrion has a double membrane.

Suggest an explanation for how this feature of the mitochondrion supports the endosymbiont theory.

1. **Inner membrane** of the mitochondrion was the cell **surface membrane** of the ingested ancient **bacterium**
2. **Outer membrane** of the mitochondrion was derived from the **cell surface membrane** of the ancestral **eukaryotic cell**
3. which formed an **endocytic vesicle**/ **food vacuole** during **endocytosis**/ **phagocytosis** of the bacterium

[3]

[Total: 14]

4 Curcumin is a yellow pigment found in the spice turmeric, which is used in curry powder. Curcumin also has medicinal properties.

In vitro experiments show that curcumin decreases the proliferation of cancer cells when cancer cells are cultured in laboratory apparatus.

In one experiment, the mitotic cell cycle of cancer cells in culture was synchronised so that they all entered G_1 phase at the same time. The cells were then transferred to culture media containing different concentrations of curcumin. Nine hours later, the percentage of cells in each phase of interphase was recorded. None of the cells had completed the G_2 phase of interphase.

The results of this experiment are shown in Fig. 4.1.

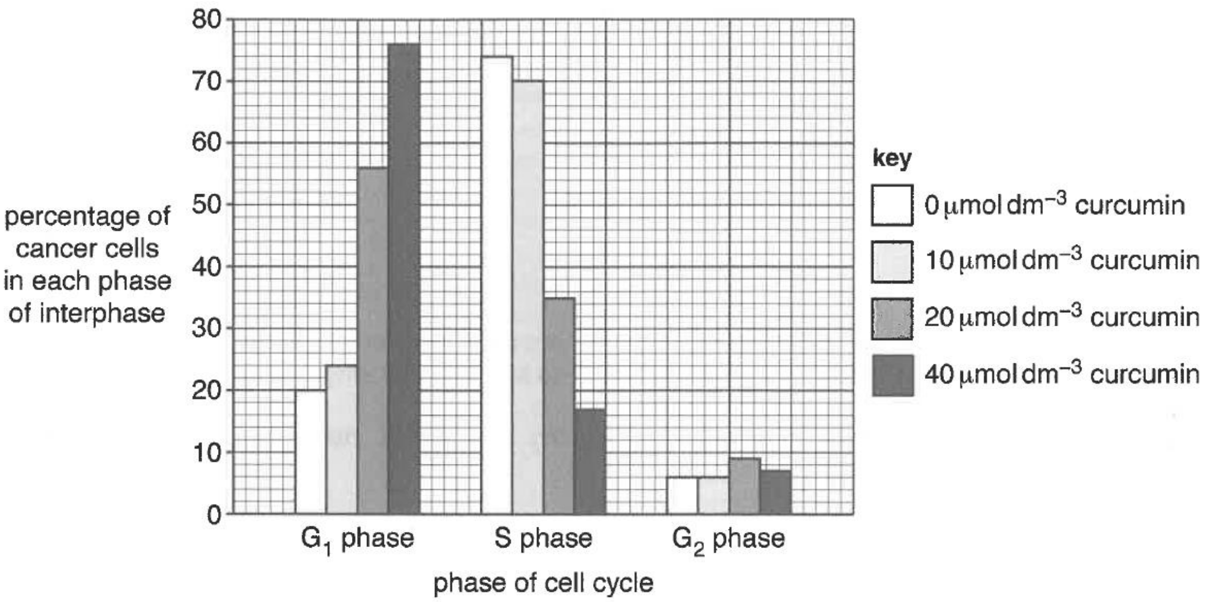

Fig. 4.1

With reference to Fig. 4.1,

(a) describe the effect of curcumin on the percentage of cancer cells in each phase of interphase nine hours after entering G_1 phase.

1. The **higher the concentration of curcumin** the cells were exposed to, the **higher the percentage of cancer cells still in the G_1 phase** nine hours later + <u>quote appropriate data to compare the absence of curcumin with a high or a range of concentrations of curcumin</u>, e.g. 76% of cells at G_1 phase in 40 μmol dm^{-3} curcumin is **higher** than 20% of cells at G_1 phase in 0 μmol dm^{-3} curcumin.
2. The **higher the concentration of curcumin** the cells were exposed to, the **lower the percentage of cancer cells in the S phase** nine hours later + <u>quote appropriate data to compare the absence of curcumin with a high or a range of concentrations of curcumin</u>, e.g. 17% of cells at S phase in 40 μmol dm^{-3} curcumin is **lower** than 74% of cells at S phase in 0 μmol dm^{-3} curcumin.
3. **No significant effect** of curcumin on the percentage of cancer cells at **G_2 phase** + <u>quote data</u>, e.g. approximately **constant** percentage (6 to 9.5%) of cancer cells were at G2 phase across all curcumin concentrations

[3]

(b) suggest how curcumin affects the mitotic cell cycle to prevent cancer cell division.

1. Curcumin **arrests/ halts/ stops** the mitotic cell cycle at the **G_1 checkpoint**
2. *(give possible mechanism for causing cell cycle arrest, e.g.)* by **causing DNA damage** to cancer cells/ acting as a chemical signal to signal the **lack of growth factors/ nutrients**

[2]

[Total: 5]

5 The white campion (*Silene latifolia)* is a flowering plant native to most of Europe, Western Asia and Northern Africa. Male plants are XY and female plants are XX.

In 1912, Baur described the first example of sex-linkage in plants, in *Silene latifolia*.

- An unusual form of *Silene latifolia* with narrow leaves was discovered, and it was found to be a male plant.
- Pollen from this plant was used to fertilise a female plant with normal broad leaves, and all the progeny (male and female) had normal broad leaves.
- When these normal broad-leaved plants were cross-fertilised, 167 plants with broad leaves and 60 with narrow leaves were produced.
- When the narrow-leaved plants were grown to maturity, all were male, while of the broad-leaved plants, 56 were male and 111 were female.

(a) Using the symbols **A** and **a** for leaf shape, draw genetic diagrams to fully illustrate the crosses above.

A: allele for broad leaves   a: allele for narrow leaves
X: represents X chromosome   Y: represents Y chromosome

Parental phenotypes: narrow leaves male X broad leaves female

Parental genotypes: X^aY X X^AX^A **[1]**

Gametes: X^a Y X X^A **[1]**

F$_1$ genotypes:

♂ \ ♀	X^A
X^a	X^AX^a
Y	X^AY

F$_1$ phenotypic ratio: All broad leaves **[1]**

<u>Cross-fertilisation of F1</u>

F$_1$ phenotypes: broad leaves male X broad leaves female

F$_1$ genotypes: X^AY X X^AX^a

Gametes: X^A Y X X^A X^a

F$_2$ genotypes:

♂ \ ♀	X^A	X^a
X^A	X^AX^A	X^AX^a
Y	X^AY	X^aY

[1]

F$_2$ phenotypic ratio: 3 broad leaves plant (2 females : 1 male) : 1 narrow leaves plant (all male) **[1]**

[5]

(b) All the narrow-leaved plants in the crosses above are male.

Describe a cross in which a female narrow-leaved plant will be produced.

1. By crossing a **narrow-leaved male plant**/ $\mathbf{X^aY}$ with a **heterozygous female/** $\mathbf{X^AX^a}$.

[1]

[Total: 6]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)**, as indicated in the question.

6 **(a)** Compare the structures and functions of tRNA and rRNA. [6]

(b) Discuss the role of proteins in the movement of substances, in biological cells **and** in multicellular organisms. [9]

[Total:15]

7 **(a)** Compare the properties and functions of embryonic stem cells and blood stem cells. [6]

(b) Discuss the significance of hydrophobic interactions in biological molecules **and** structures. [9]

[Total:15]

6 (a) Compare the structures and functions of tRNA and rRNA. [6]

STRUCTURE

[Similarities]
1. Both contain **uracil** instead of thymine.
2. Both are held together by **phosphodiester bonds**.
3. Both contain **secondary structures**/ **complementary regions** held by **hydrogen bonds**.
4. Both are **single-stranded**.

[max 2]

[Differences]
5. tRNA has a **clover-leaf** shape, but rRNA has a **spherical/ globular** shape.
6. tRNA has an **anti-codon** loop, but rRNA does not.

[max 1]

FUNCTION

[Similarities]
7. Both are involved in **translation** to synthesise polypeptides.

[max 1]

[Differences]
8. **tRNA carries a specific amino acid** to the **A site** of the **ribosome**
9. due to **complementary base-pairing** between its **anticodon** and the **codon** on mRNA *(but)*
10. rRNA is a **structural component** of the **ribosome**/ **associates with ribosomal proteins** to form **ribosome**/ **ribosomal subunits**
11. part of rRNA contains the ribozyme **peptidyl transferase**, which catalyses the formation of **peptide bonds**

[max 3]

6 (b) Discuss the role of proteins in the movement of substances, in biological cells **and** in multicellular organisms. [9]

[Transport proteins]
1. **channel protein** provides a **hydrophilic channel/ pore/ pathway**
2. for the movement of small **ions**/ **charged** substances across a membrane
3. by **facilitated diffusion**

4. **carrier protein** undergoes **conformational change**
5. to transport (usually) **polar** molecules/ named example, e.g. glucose
6. involved in **facilitated diffusion** *(award once only)*/ **active transport**
7. *ref. to* specific example, e.g. sodium-potassium pump

8. transport by channel/ carrier protein is **specific**
9. allows movement of charged/ polar substances across **hydrophobic core**/ **fatty acid tail** layer of membrane

10. **aquaporins** are involved in the transport/ facilitated diffusion of **water** molecules across a membrane

[Receptor-mediated endocytosis]
11. **receptor** proteins bind to specific ligands/ molecules for uptake via **receptor-mediated endocytosis**
12. *ref. to* **clathrin**-coated pits

[Microtubules]
13. network of **microtubules** allow movement of **vesicles**
14. *ref. to* vesicles carried by **motor proteins** along microtubules
15. microtubules also aid in the **separation** of (sister) **chromatids** during **mitosis**/ **meiosis**

[Haemoglobin]
16. **haemoglobin** involved in **transporting oxygen** around the body
17. *ref. to* quaternary structure of haemoglobin which contains **four prosthetic haem groups**
18. *ref. to* **co-operative binding of oxygen** by haemoglobin
19. one haemoglobin molecule has the ability to transport **four oxygen molecules**, hence increasing efficiency of transport *(idea of)*

20. AVP, e.g. electron transport chain on inner mitochondrial membrane/ thylakoid membrane

21. QWC: At least 2 MPs each from 1 – 15, and 16 – 19

7 (a) Compare the properties and functions of embryonic stem cells and blood stem cells. [6]

PROPERTY

[Similarities]
1. Both are capable of **long term self-renewal** via **mitosis**/ can **divide continually**.
2. Both are **unspecialised/ undifferentiated**.
3. Both have the **ability to differentiate** to subsequently give rise to specialised cells under suitable conditions.

[max 3]

[Differences]
4. Embryonic stem cells are **pluripotent** but blood stem cells are **multipotent**.
5. Embryonic stem cells are able to **differentiate into almost any cell type** that makes up an organism **except those of the extra-embryonic tissue**, but blood stem cells can only differentiate to give **cells that form the blood tissue** (limited range of cell types).
6. Embryonic stem cells are found in the **inner cell mass of the blastocyst**, but blood stem cells are found in the **bone marrow**.

[max 2]

FUNCTION

[Differences]
7. Embryonic stem cells divide to **form the three germ layers**/ **endoderm, mesoderm and ectoderm**
8. but blood stem cells give rise to blood cells that **replace worn out/ damaged blood cells**.

[max 2]

7 (b) Discuss the significance of hydrophobic interactions in biological molecules **and** structures. [9]

[Proteins]
1. hydrophobic interactions are formed between **non-polar R groups**/ **amino acid** residues
2. contribute to specific **3D conformation**/ **tertiary structure** of proteins
3. *ref. to* **<u>specific</u> active sites**/ **binding sites** that result

[DNA]
4. hydrophobic interactions are formed between nitrogenous **bases** within **DNA** molecule
5. **stabilises double-helical** structure

[Haemoglobin]
6. **haem** group has a hydrophobic **porphyrin ring**
7. **hydrophobic clefts** formed on each of the four **subunits** of haemoglobin allow 4 haem groups to be inserted
8. *ref. to* function of haemoglobin in **oxygen transport**

[Triglycerides]
9. hydrophobic interactions form among **fatty acid tails/ chains** of **triglycerides**
10. *ref. to* triglycerides being **insoluble <u>in water</u>**
11. allowing them to be **stored in large amounts** as oil/ lipid droplets in cells
12. *ref. to* **<u>energy</u> storage** function of triglycerides in animals

[Membranes]
13. **phospholipid bilayers** are stabilised by hydrophobic interactions among **non-polar fatty acid tails/ hydrocarbon chains** of phospholipids
14. these weak interactions allow the membrane to **remain fluid**/ be in **constant motion**/ allow phospholipids to **move laterally** within monolayer
15. *ref. to* hydrophobic core of membranes allowing **selective movement of substances** across/ allowing **only uncharged/ non-polar substances to freely diffuse** across
16. hydrophobic interactions also allow **membrane proteins** to be <u>held in place</u>
17. *ref. to* hydrophobic interactions formed between **non-polar amino acid** residues on <u>surface</u> of membrane protein and **non-polar fatty acid tails/ hydrocarbon chains** of phospholipids

18. AVP

19. QWC: At least 2 MPs each from 1 – 12, and 13 – 17